Income Variables and Sources in the Canadian Community Health Survey Share File

1. Introduction

In anticipation of linking to tax records, the 2016 income questions were not asked in the majority of cases. The income questions were only asked where respondents refused the linkage agreement or where there was a low probability of finding a link (based on preliminary work). While many respondents permitted the link to tax records and also agreed to share their data, restrictions were imposed that prohibited the sharing of tax data to share partners; in compliance with the Income Tax Act. This document describes the income variables on the 2016 share file and how they were populated.

2. Respondent Data

The income variable most commonly used in analyses is total household income (variable INC_021). In 2016, about one quarter of respondents provided an estimate of the annual household income from 2015. The response rate was similar for those eligible to respond to the total personal income question (INC_041). Another ~5% of respondents provided an estimated range of their household income and ~3.5% provided a range of their total personal income. Respondent data was used wherever possible to populate the share income variables.

3. Income imputation

Where respondents provided an income range for total household or total personal income, donor income imputation was performed. Using characteristics of the respondent and their household, a nearest neighbour donor was found within the provided income range. The modeled household and personal income defined in the following section is used as a distance measure to determine which donor-recipient pair is the "nearest" within imputation classes.

3.1 Modeled Income for Donor Imputation

An important auxiliary variable used in the imputation process is modeled income. It is computed for all CCHS respondents (whether they responded to the income questions or not) using a process similar to the modeling described in section 4, it is done at both the household and personal levels.

Using a regression model, household or personal income is predicted based on household size, education of household members, location information, employment and health characteristics as well as ages and sex of household members and the median household income from T1 tax data for the postal code the respondent resided in. These models differed in both the variables and the coefficients used to predict income at the personal and household levels.

3.2 Choosing a Donor

Imputation classes are formed differently depending on whether the income being imputed is a total household income or total personal income. For total household income, imputation classes are based on the income range and household size. For total personal income, imputation classes are based on age and sex in lieu of the household size. The imputation was also constrained by the income range provided by the respondent as well the total household or personal income if provided.

4. Imputing from an Income Model using CCHS data and Admin Records

For the remaining three quarters of respondents, the donor imputation method used above is inappropriate due to the large number of records requiring an imputed value compared to the number of potential donor records. For this reason, income models were created using aggregate tax data and a number of CCHS variables.

The CCHS sample was divided into 10 groups, each group having a representation of all incomes. To impute the incomes for one group, the remaining nine groups of CCHS respondents were collapsed together and a model was fit based on their aggregated tax data. The CCHS variables, along with the log of median income in the postal code (or just the first three characters of the postal code for small postal codes), were used as predictors for income. The model found was then applied to the remaining group to impute an income for each respondent

The CCHS variables used in modelling were:

- DHHDVHSZ (Household size)
- DHHDVL18 (Number of household members less than 18)
- Geodvur & geodvur2 (Population center or rural area)
- ehg2dvh9 (Highest education in household)
- ehg2dvr9 (Highest education for respondent)
- GEO_PRV (Province/territory)
- LBF_10 (Employee, Self-employed, family business)
- LBF_030 (Had more than one job)
- LBFDVHPW (Total usual hours worked per week)
- LBFDVPFT (Full time, part time status)
- LBFDVING (Industry group, based on NAICS 2012)
- LBFDVOCG (Occupation group, based on NOCS 2011)
- SDC_015 (Identify as, First Nations, Métis or Inuk (Inuit))
- SDC_010A-SDC_010U (Ethnic/cultural origin)
- SDC_IM3 (Landed immigrant)
- DHH_OWN (Own or rent dwelling)
- DHH_MS (Marital status)
- DHH_AGE (Age of respondent)
- DHH_SEX (Sex of respondent)
- GEN_005 (Self perceived health)

- GEN_015 (Self perceived mental health)
- GEODVPSZ (Population center or rural area classification)
- FSCDVHFS (Household food security status)
- MAC_005 (Main activity last week)
- INS_010A (Government sponsored insurance plan)

The model used took the form of:

$$\log(y) = \beta_1 X_1 + \beta_2 X_2 + \cdots + \beta_n X_n$$

Where y was the income from tax files, and β_i are the coefficients and X_i are the values of the independent variables that were retained from the list above. The predicted values from the models were converted to household and personal incomes and used to create the final 2016 CCHS share income variables.

5. Data File

Roughly two thirds of the 2016 CCHS share respondents have their total household income (or personal income, where applicable) imputed from a model. The quality of the modelled incomes was assessed prior to dissemination of the data. While the models generally determined if respondents had higher or lower incomes, the modelled incomes lacked the precision implied with an exact dollar amount. For this reason, the variables for total income at the household and personal level (INC_021 and INC_041) are not on the share file. They have been replaced by grouped variables INC_020 and INC_040. The income group variable on the CCHS share file correctly assign respondents to their income group in about 60% of cases. Respondents are within one income group in about 88% of cases. Likewise, the income decile variables (INCDVRCA, INDVRPR and INCDVRRS) have been replaced by quintile variables (INCDVSCA, INCDVSPR, INCDVSRS). These quintiles are calculated based on the exact dollar amount provided by the respondent or imputed from a CCHS donor or from the model. The sources of income variables (INC_005A-O, INC_Q025A-O), and main source of income variables (INC_015, INC_035) have been removed due to the high missing rate with only respondent data as a source. The flag INCFSH and INCFSP inform users of the method used to populate the income variables.

Table 1. Values for INCFSH

INCFSH	2016 Code
1	Reported by respondent
2	Imputed from a donor
3	Imputed from an income model

Table 2. Values for INCFSP

INCFSP	2016 Code
1	Reported by respondent
2	Imputed from a donor
3	Imputed from an income model
6	Not applicable

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